BIG DATA, INDIVIDUAL ASSIGNMENT 2

Faster and Sparse Matrix Multiplication

Optimised dense methods and sparse matrices in Java

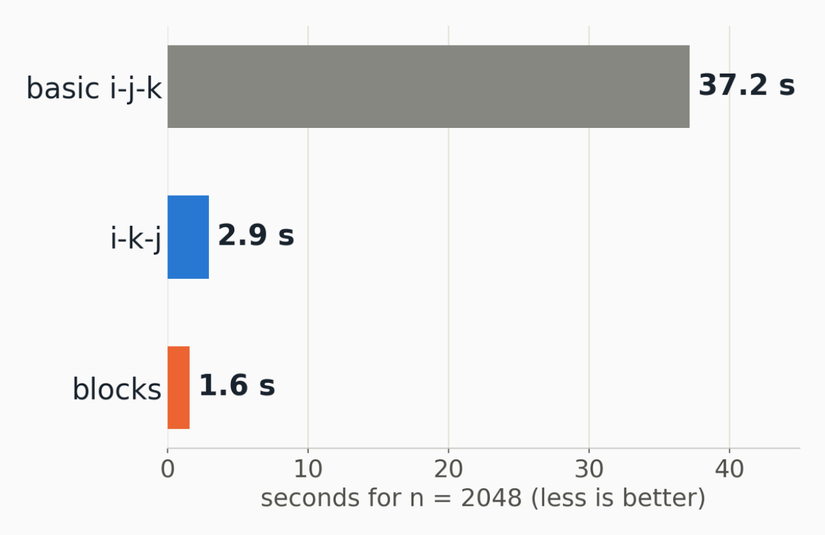
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Grado en Ciencia e Ingeniería de Datos, 2026-2027

Part A: making normal multiplication faster

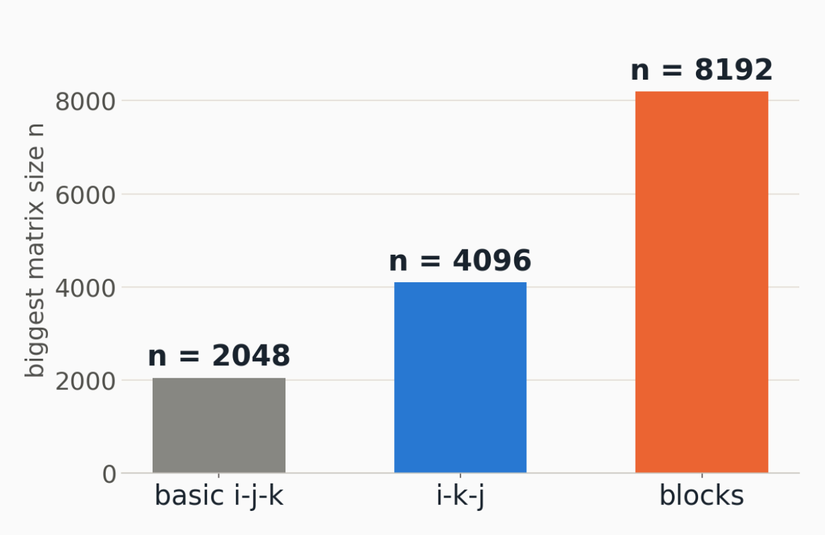
Same operations, same result. Only the order of reading the memory changes: basic i-j-k, i-k-j, and blocks.

Time for a 2048 × 2048 matrix



Blocks are 24 times faster than the basic version.

Biggest size in 20 minutes

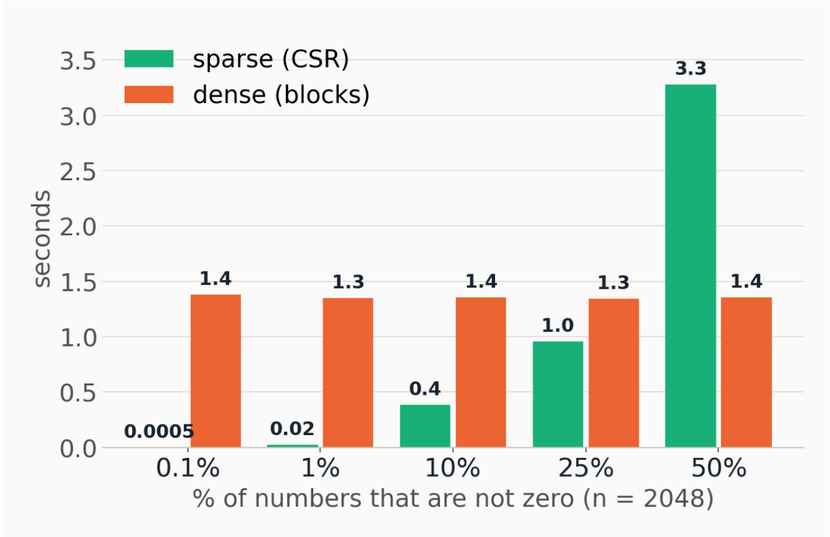


With blocks I could multiply matrices 4 times bigger.

Part B: sparse matrices, and conclusions

Sparse = almost all numbers are zero. CSR saves and multiplies only the numbers that are not zero.

Sparse (CSR) vs normal, n = 2048



Normal: always about 1.4 s. CSR: very fast with few nonzeros.

× 24 — Blocks: 24 times faster than the basic version.

× 2,800 — CSR with 0.1% nonzeros: thousands of times faster.

50% — With many nonzeros, normal is better.

1% → 18% — The result C gets full of numbers very fast.

Code, data and report: github.com/Pablo5250/matrix-multiplication